Ascend Super-Operator Fusion

One kernel per phase, not thirteen

Zhipu AI · GLM-5 on Huawei Ascend

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Kernel Fusion & Runtime · arxiv.org/abs/2602.15763

GLM-5 ships on domestic NPUs from day one

- **744B** params, **40B** active, **256** experts, **80** layers, DSA sparse attention
- Adapted to **7** Chinese chip platforms; Ascend Atlas is the case study
- Target: the whole model served from **one** Atlas 800T A3 machine

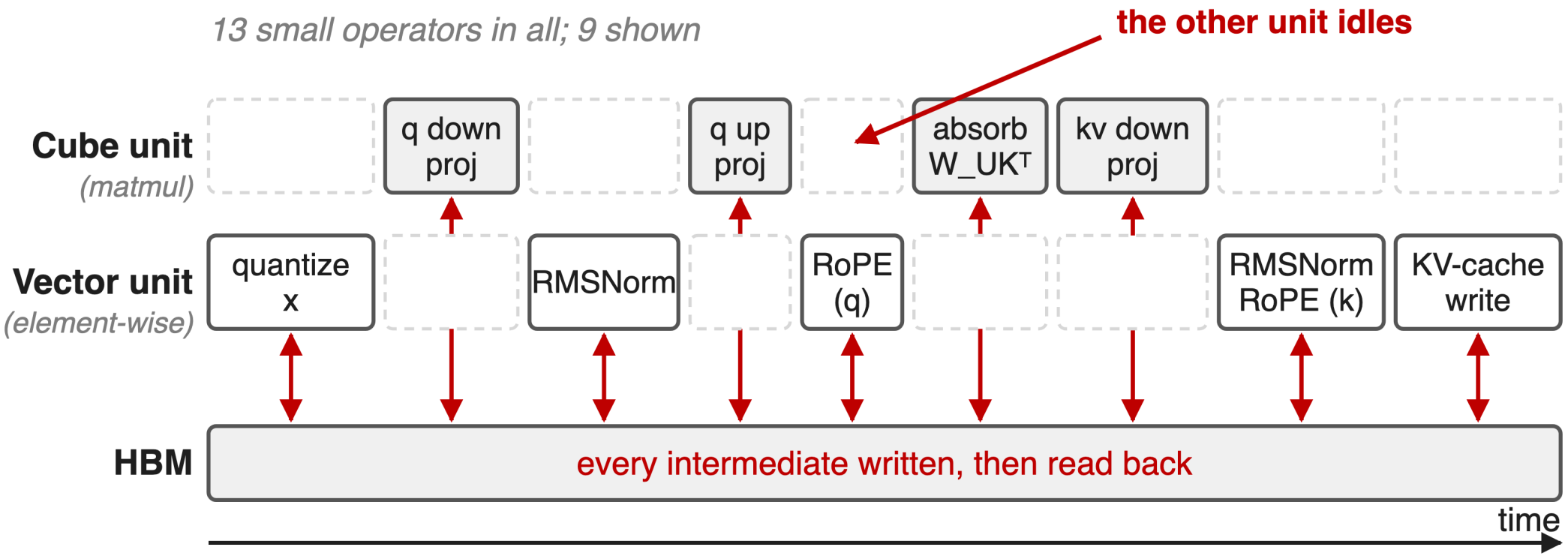
A thousand tiny kernels per decoded token

≈1,000

kernels per decoded token for MLA pre-processing alone — 13 small ops in each of 80 layers

Derived from GLM-5 report (arXiv 2602.15763): 13 pre-processing ops, §5 p.21-22; 80 layers, §2.1 p.4

Op-by-op execution idles half of the AI core



Each op uses one unit, writes its result to HBM, and the next op reads it back

Insight: the operator boundary is the cost

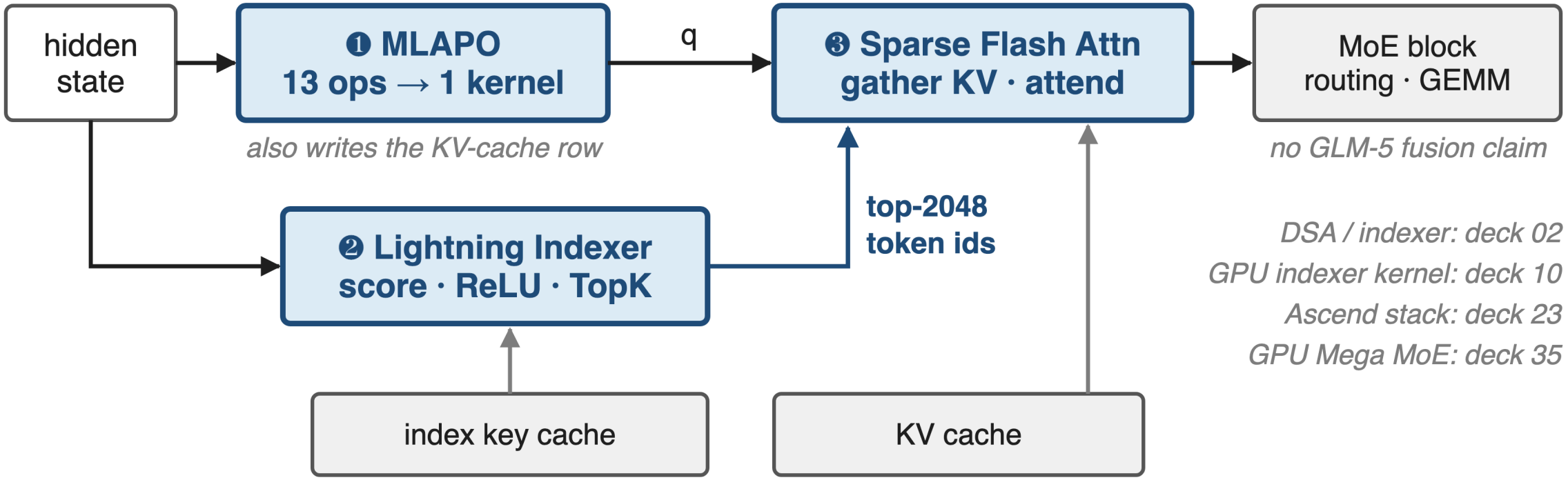
- Every tensor between two ops is live for microseconds and read once
- The Cube and Vector units are independent hardware, not a single pipeline

Insight: the operator boundary is the cost

- Every tensor between two ops is live for microseconds and read once
- The Cube and Vector units are independent hardware, not a single pipeline

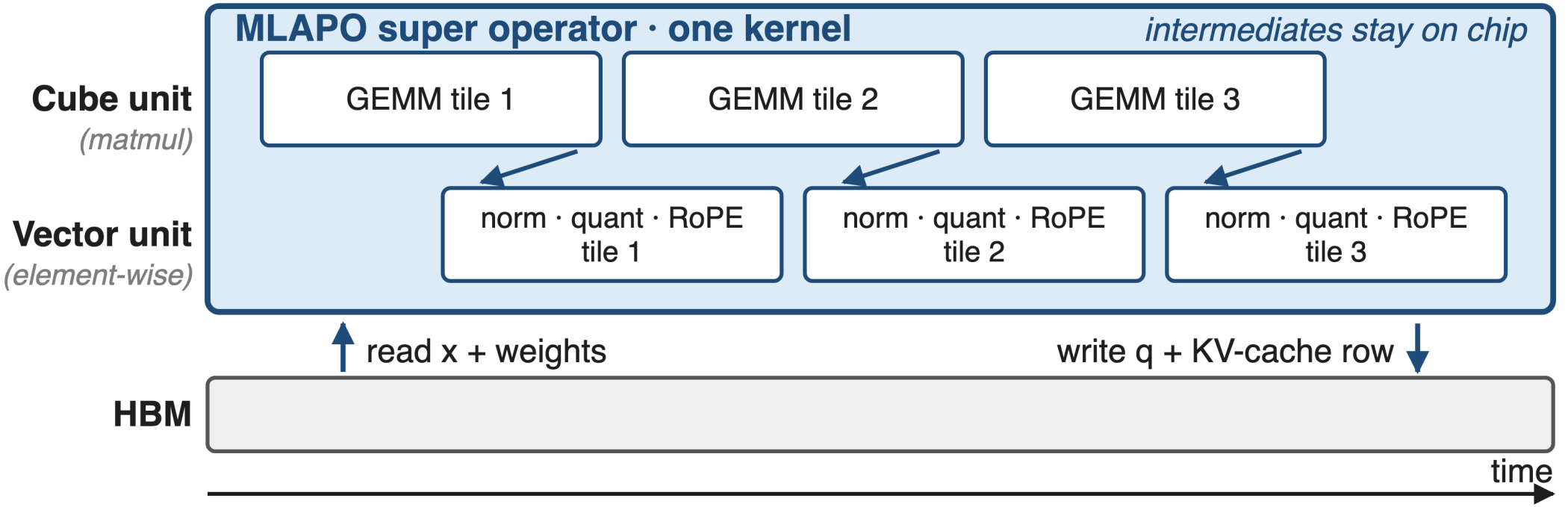
Fuse the phase: intermediates stay on chip, and both units work at the same time

Three super-operators carry the decode path



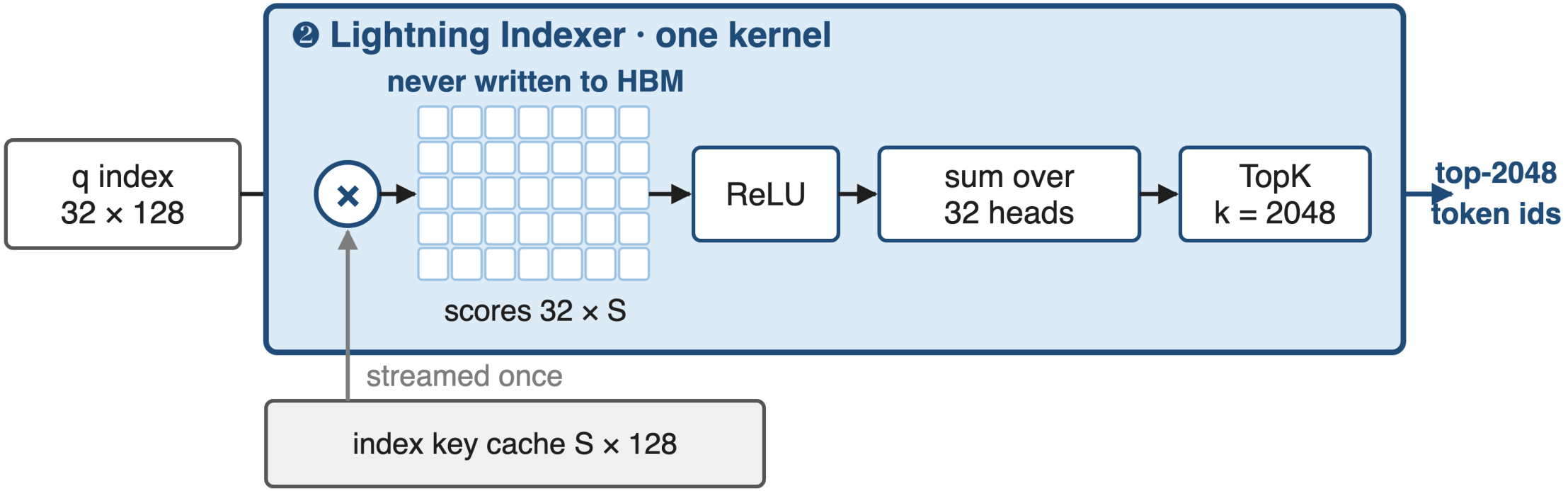
GLM-5 report §5: Lightning Indexer, Sparse Flash Attention, MLAPO. The MoE block is not part of its fusion claims.

❶ MLAPO fuses 13 pre-processing ops



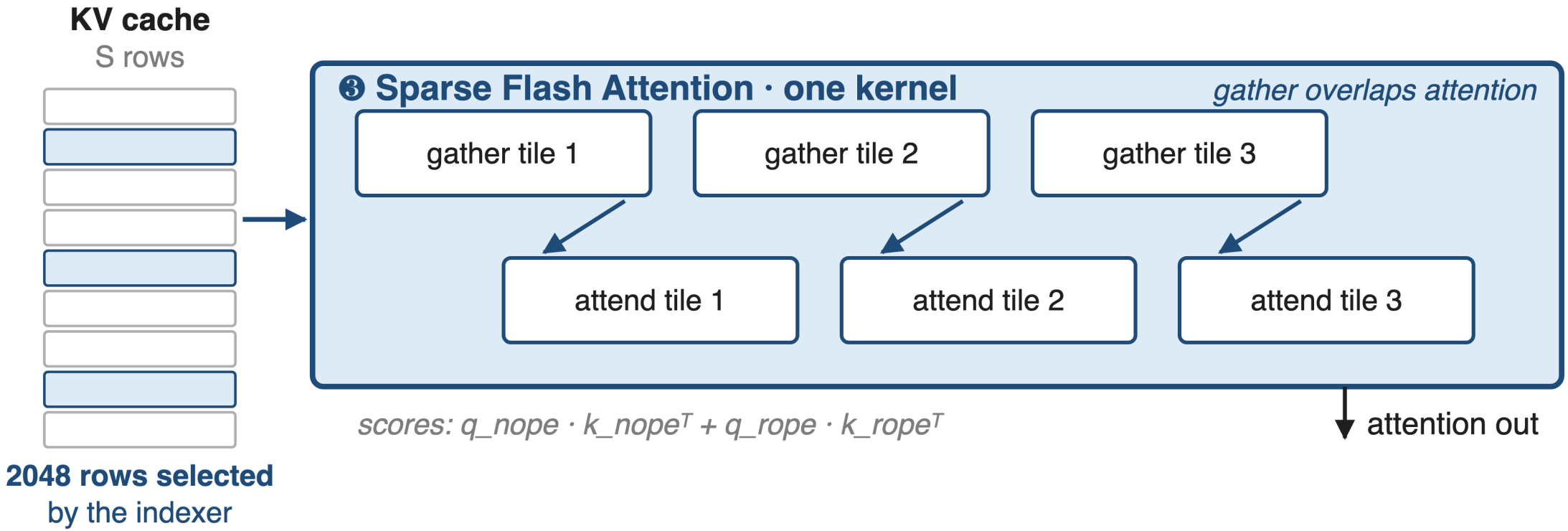
Quantize, project, normalize, rotate, absorb, write cache — one kernel, two units

❷ One kernel scores, rectifies and ranks



The 32 × S score matrix never becomes a tensor in HBM

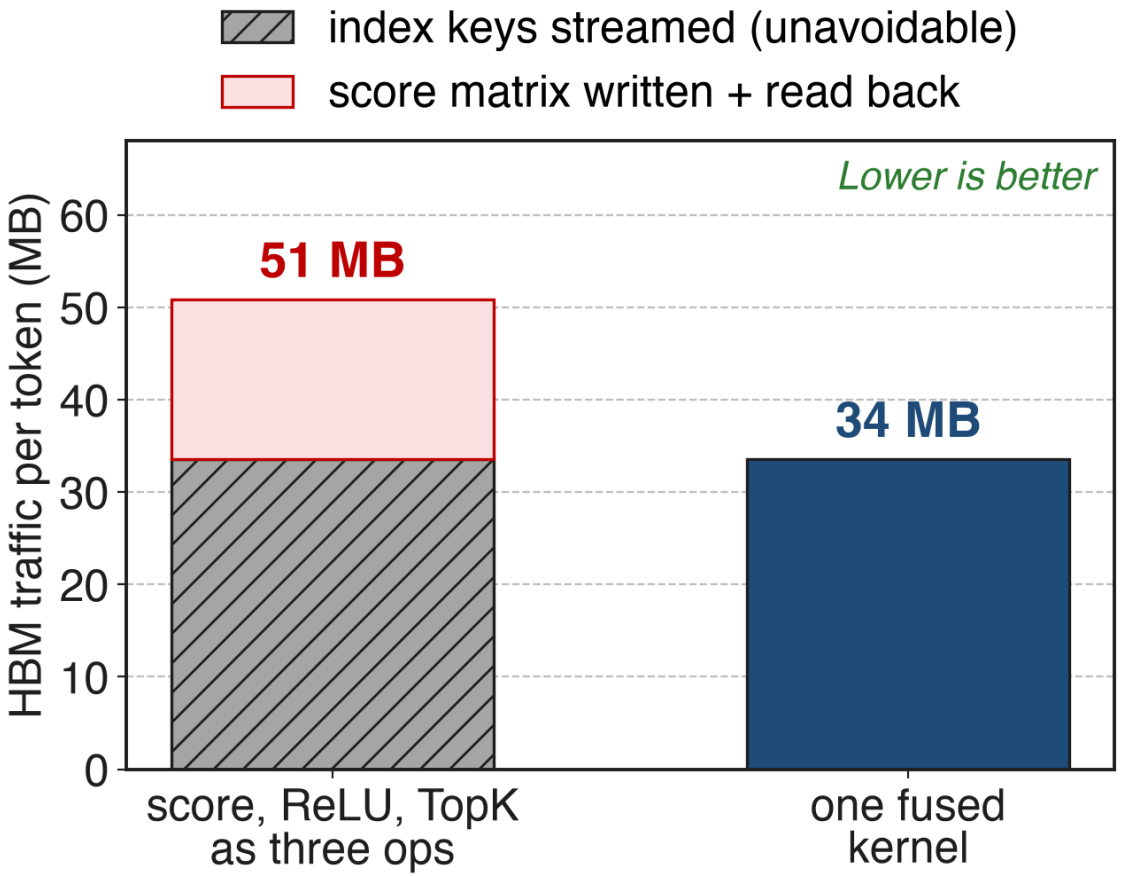
❸ Gather and attend in one kernel



Gathering the next tile of KV overlaps the attention math of the current one

Fusion removes ~17 MB per decoded token

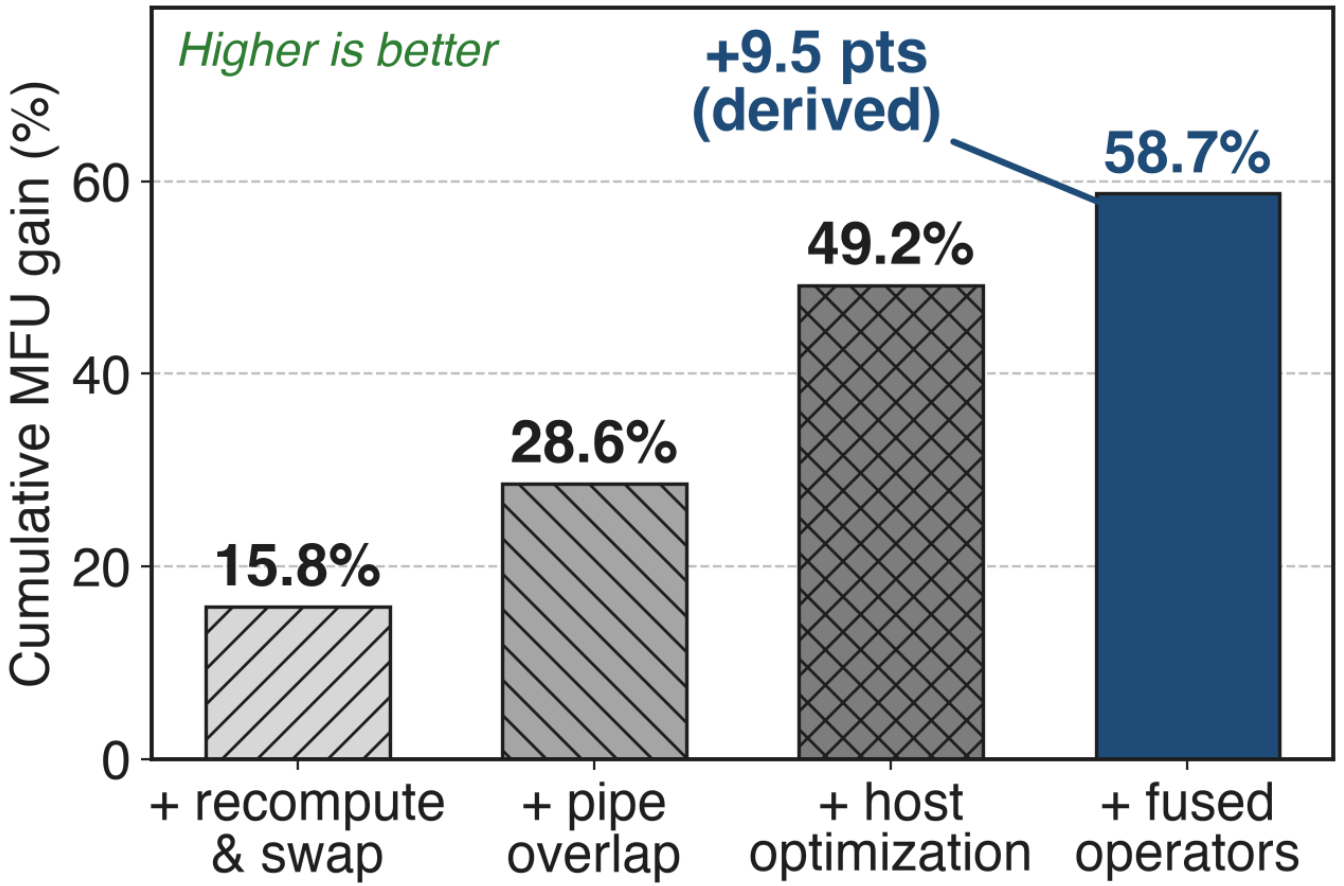
- Index keys must be streamed either way
- The score matrix is pure fusion profit
- **1.5×** less indexer traffic per token



Analytic, not measured: bf16, 128K context, 32 index heads of dim 128 (Table 10)

Fused operators: the last +9.5 MFU points

- Cumulative gain over the baseline
- Final MFU **30.0%**
- Fusion comes last, after recompute, overlap and host fixes



Pangu Ultra MoE (arXiv 2505.04519) Table 5: 718B MoE training on 6K Ascend NPUs — different model, training not inference

One domestic node, half the long-context cost

50%

lower deployment cost in long-sequence scenarios; a single Chinese node reported comparable to "dual-GPU international clusters"

GLM-5 report §5 p.22 — attributed to all hardware-level co-optimizations together (quantization + fusion + engine)

The cost: hand-written kernels, frozen shapes

- **One kernel per phase**, written against one model's shapes
- **Tied to NPU generations**: A2/A3 vs Ascend 950 support differs

- No per-kernel numbers published
- Change the architecture → rewrite the kernel
- MoE fusion is stack-level, not a GLM-5 claim

Summary

- MLA pre-processing alone: **13** ops per layer, **≈1,000** kernels per decoded token (derived)
- **Insight**: intermediates exist only because the operator boundary does
- Three super-operators: MLAPO, Lightning Indexer, Sparse Flash Attention
- **1.5×** less indexer traffic (analytic); **50%** lower long-context deployment cost (whole stack)

https://arxiv.org/abs/2602.15763